
Digital Essence

An evaluation of the Configuration Table from the Essence
methodology

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Abstract:

This report evaluates the Configuration Table from the Essence methodology as a visual inquiry tool, following the Design Theory of Visual Inquiry Tools. We adopt Naur's definition of Programming as Theory Building and Schön's concept of Designing as a Reflective Conversation, in addition to CASE tool theory, to determine essential features for a digital Configuration Table. We introduce the first version of Digital Essence, including its user interface and the technologies used in its development. We propose an experiment to gather data on Digital Essence and answer the question of how practitioners use the Configuration Table. Finally, we discuss the current state of Digital Essence, outline potential future work, and offer concluding remarks on fulfilling the research questions.

The content of this report is freely available, but publication (with reference) may only be pursued due to agreement with the author.

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Preface

A special thanks to our supervisor Ivan Aaen, for providing useful insights into the foundation of the Essence Methodology.

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1 | Introduction

In software development, understanding a problem, how it changes, and the users interacting with it is essential for providing an optimal solution. Agile methodologies like Scrum and XP can assist in developing software as solutions to problems under the conditions that problems change. However, these methodologies focus heavily on the development processes. The Essence methodology is an approach to managing software innovation projects, where idea generation and its maturation are in focus. The Essence methodology addresses the uncertainty and provides structures to manage and understand potential changes in the problem space[1].

In this project, we focus on one of the structures in Essence, the Configuration Table, which is used to discover and keep relevant "ground truth" information about the innovation project.

1.1 Motivation

The Essence methodology for innovation projects is grounded in Deweyan pragmatism and focuses on facilitating the maturation of ideas through joint inquiry and interaction in problematic situations. As part of Essence, the Configuration Table serves as a critical tool and "ground truth" for processes such as RST Reviews¹. It is intended to represent and create new knowledge about a problematic situation, with each cell representing a question that can lead to further knowledge generation. In other words, it serves as the central repository for understanding the problem. The nature of the Configuration Table is dynamic in that it reflects our current understanding of a problematic situation and should thus be continuously updated throughout the lifetime of a project.

However, the current Configuration Table in Essence methodology is limited by its inability to represent the complex relationships between the cells. These relationships are central to our understanding of the problem, but the current format must allow for clear visualization. As a result, the Configuration Table can be challenging to comprehend, and proper reviews may be compromised, as changes in one cell may necessitate corresponding changes in other cells based on the established relationships.

¹A review of the Rationale, Strategic and Tactical levels of the Configuration Table

As the Configuration Table aims to facilitate joint inquiry into a given problem and solution space, it is interesting to analyze if the table can be defined as a Visual Inquiry Tool (VIT), as these tools serve the purpose of creating a conversation regarding a specific point of interest. To support this analysis, the design theory of visual inquiry tools presented in [3] can be used, where authors propose principles for designing new tools and evaluating existing ones. Furthermore, the Configuration Table needs to suggest or consider what medium in which to use the Configuration Table. In this project, we propose a digital version of the Configuration Table as a means of online collaboration of software innovation projects. With a digital version of the Configuration Table, we can explore how it is used and what structures in the Configuration table are essential for effective joint inquiry.

In this paper, we seek to answer the following questions:

- RQ1: Is the Configuration Table in alignment with the Visual Inquiry Tool Design Theory Principles[3]?
- RQ2: Which features are essential for a digital version of the Configuration Table?
- RQ3: How is the digital version of the Configuration Table used?

To answer the research questions, we will analyze the Configuration Table using principles of the Visual Inquiry Tool (VIT) Design Theory [3] to determine how well it can be described as a visual inquiry tool. Additionally, we will use computer-aided software engineering tool theory [7] to determine the best methodology support philosophy for a digital version of the Configuration Table. From this philosophy, we will identify the essential features that align with it and test these features through experiments to see if they are used by practitioners as intended. Additionally, we will conduct interviews to explore any unevaluated features deemed necessary by practitioners. Finally, we will use data from the experiments to understand how practitioners use the digital Configuration Table and if its proposed format supports their ways of using it.

1.2 Related works

At the time of writing, some works on improving the structures in the Essence methodology exist. In [6], the author considers Essence RST reviews as a visual inquiry tool and evaluates it based on VIT principles. Furthermore, this work also evaluates the Configuration Table based on VIT principles, and the author suggests support to help fill out the table. Another related work to ours, [5], considers the practical utility of Essence and aims to evaluate its practical use of it through a reflective process. Here the author concluded that filling out the Configuration Table worked well when reflecting on the connection between cells. However, according to the author, the table has some drawbacks. One is the need for more understanding of the terminology and guidance for its use.

1.3 Report structure

In this report, chapters 2 and 3 establish a theoretical foundation of Essence and the Visual Inquiry Tool Design Theory. Chapter 4 comprises an analysis of the Configuration Table using the VIT Principles. In chapter 5, system design considerations for developing a digital version of the Configuration Table are described based on the theory of Computer-aided Software Engineering (CASE) tools. Chapter 6 discusses theory building, incorporating Naurs' definition of Programming as Theory Building and Schön's concept of Designing as a Reflective Conversation. Chapter 7 presents the first version of Digital Essence, including its user interface and the technologies used in its development. In chapter 8, we propose an experiment to gather data on the first version of Digital Essence, along with a description of the data collection and storage methods. The report is concluded with a Discussion and Future work in chapter 9 and a conclusion in chapter 10.

We will be assessing this project based on (1) how well we answer RQ1, (2) the appropriateness of the essential features selected for RQ2, and (3) how suitable our data collection methods are to answer RQ2 and RQ3 based on the planned experiments for the next semester. The evaluation will also consider the technology choices, their impact on the project, and the results of following Naur's and Schön's concepts.

2 | Essence Methodology

This chapter aims to provide a comprehensive overview of the central elements of the Essence methodology. Specifically, we outline the critical components of Essence, which include Views, Roles, Prospects, the Configuration Table, and Reviews. By detailing these elements, this chapter serves as the theoretical foundation for our research's subsequent investigation of Essence.

2.1 Essence Views

Views, in Essence, provide perspectives on the particular and the whole of a project while maintaining a systematic approach to the problem. It helps provide new view-points when working on a project, specifically an innovation project. These view-points welcome new ideas and aspects into the project, enabling a creative way of working on the problem[1, Foreword, p. xx]. The following views are presented in Essence:

Situation Is the view regarding understanding the problem's circumstances and describing general information about the problem and how it is perceived across the team. It also identifies the resources available to work on the problem and the solution. All this contributes to how the situation is developed and helps to understand specific parts of a problem.

Contribution Is what we design and do. The contribution is the constructs we devise as part of the solution. It involves the designs, features, experiments, partial validation of resources available in the project, and the idea and design that solves the problem.

Solution Managing the project is an ongoing conversation between the problem and the solution. Our perception may change while discovering additions to the problem. Does our contribution provide a meaningful solution to the problem? In the solution view, we take stock of the state of the project. Does our current contribution contribute to the current understanding and solution to the problem? In the solution view, we see what limitations our solution has and prioritizes with stakeholders.

Valuation Ensuring the project is on the right track. How well did the solution solve the problem? Assess how the current way of thinking fits the problem. We

want to look into what effect our solution has on the problem. The valuation view requires evidence and appraisal.

2.2 Essence Roles

Essence introduces four roles that encourage team members to take on diverse responsibilities. The roles are created to inspire diverse thinking, based on the Essence Views[1, p. 47].

Child Any team member that explores the situation. The child is creative and offers ideas and proposals without responsibilities. This role is fleeting, and any team member can have the Child role.

Responder Any developer that is working on the contribution. The Responder role provides technological responses to challenges. Always with a view on technological options.

Challenger Any team member that provides the reasoning and politics underlying the solution. Aiming for a solution with a good balance between problem and contribution. Open to new ideas and sound arguments.

Anchor The team member in charge of valuation. The Anchor role is a team captain facilitating credible evaluations and providing a reasonable basis for pivoting or persevering. This role is *not* a manager.

2.3 Essence Prospects

In Essence, a prospect denotes the contribution which we believe will solve the problem if put to proper use. A prospect reflects an understanding of the problem and an adequate path to solving it. The term is named 'prospects' to reflect the mutable nature of both problem and solution, indicating that our ideas of the problem and how to solve it might change over time. A prospect is the focus of a project, provided that it holds to our understanding of the problem and solution. If the focus of the project changes, a new prospect must be selected[1, p. 92-93, 105–110].

2.4 Essence Configuration Table

The Configuration Table is a collection of statements related to a specific understanding of the problem and how to solve it (problem and prospect). The Configuration Table aims to establish an overview from opposing viewpoints and maintain consistency in a project[1, p. 118]. Table 2.1 is an illustration of the Configuration Table, divided into four columns corresponding to the four views of Essence, and the three rows: **Rationale**, **Strategy** and **Tactics**.

View	Situation (Reflection)	Contribution (Transaction)	Solution (Reasoning)	Valuation (Appreciation)
Rationale (Why?)	Problematic • Challenge • Problem (Grounds)	Leverage	Resolution • Prospect (Claim) • Warrant • Backing	Criteria for Resolution
Strategy (What?)	Outer Environment	Inner Environment	Qualification • Qualifier • Rebutal	Criteria for Inner Environment
Tactic (How?)	Manifestations	Features	Value • Proposition	Criteria for Value Proposition

Table 2.1: Configuration Table

Elements in the Configuration Table are inspired by Stephen Toulmin’s argumentation model[17], which breaks arguments down into six components. The **claim**, which is an assertion or statement we want to argue, holds. The **grounds** is the facts that support the claim. The **warrant** is the assumption we make to link the grounds and the claim, and we have **backing** to support the argument’s warrant. As for limiting the argument, we have a **qualifier**, and a justification for that qualifier is the **rebuttal**. These components are colorized by yellow and green in Table 2.1, and resides within the **problematic**, **resolution** and **qualification**

2.4.1 Rationale

The rationale level focuses on the *why* for a given prospect. It contains the reasoning and logical basis for the course of action. It is described in terms of [1, p. 109]: *The Problematic*; a challenge and a problem within, *The Leverage*; the keystone of the Inner Environment which can be utilized to build a great solution, *The Resolution*; a prospect, why solving the problem is essential, and why the prospect is a valuable solution and *The Criteria for Resolution*; the basis of which the Resolution should be evaluated.

2.4.2 Strategy

The strategic level focuses on the *what*. It explores the internal and external surroundings of the project and is described in terms of the following: *Outer Environment*; “the surroundings in which our problem resides, and where our contribution will operate”[1, p. 109], *Inner Environment*; “the substance and organization of what we are designing and implementing.”[1, p. 109], *Qualification*; the shortcomings of the solution and why these are acceptable and *Criteria for Inner Environment*; the basis on which the Inner Environment should be evaluated.

2.4.3 Tactics

The tactical level focuses on the *how*. It details how specific ends can be achieved and is described in terms of [1, p. 10]. *Manifestations*; events, actions, or objects occurring in the Outer Environment, to be handled by the system, *Features*; properties of the system, based on the Inner Environment. Used to provide Value Propositions, *Value Propositions*; desired goals that partially solve the problem, and *Criteria for Value Propositions*; the basis on which the Value Propositions should be evaluated.

2.5 Patterns in the Configuration Table

The cells of the Configuration Table are interconnected both vertically and horizontally. Figure 2.1 illustrates the different patterns within the Configuration Table.

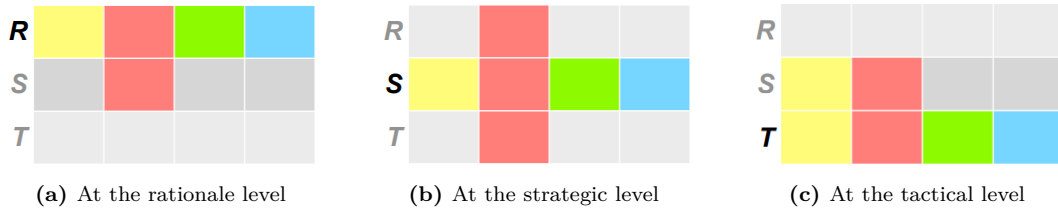


Figure 2.1: Illustration of the patterns in the Configuration Table[1]

At the rationale level

Horizontally, the Prospect is related to the problem. The warrant justifies why the problem is worth solving, and the backing supplies the solution's attractiveness. The Prospect is linked to the Leverage Points, elements that can be utilized in the solution. *Vertically*, the Leverage Points are supported by the Inner Environment[1, p. 116].

At the strategic level

Horizontally, the Inner Environment is related to Outer Environment, and the Qualification describes the limitations. *Vertically*, the Inner Environment supports the Features and creates value through the Leverage Points[1, p. 113].

At the tactical level

Horizontally, the Manifestations should be handled by the system using Features which is part of providing or achieving a Value Proposition. *Vertically*, the Manifestations are based on the Outer Environment, and the Features are implemented via the Inner Environment[1, p. 110].

Relations summarized

The abovementioned relations are summarized in the following illustration of the Configuration Table in Figure 2.2.

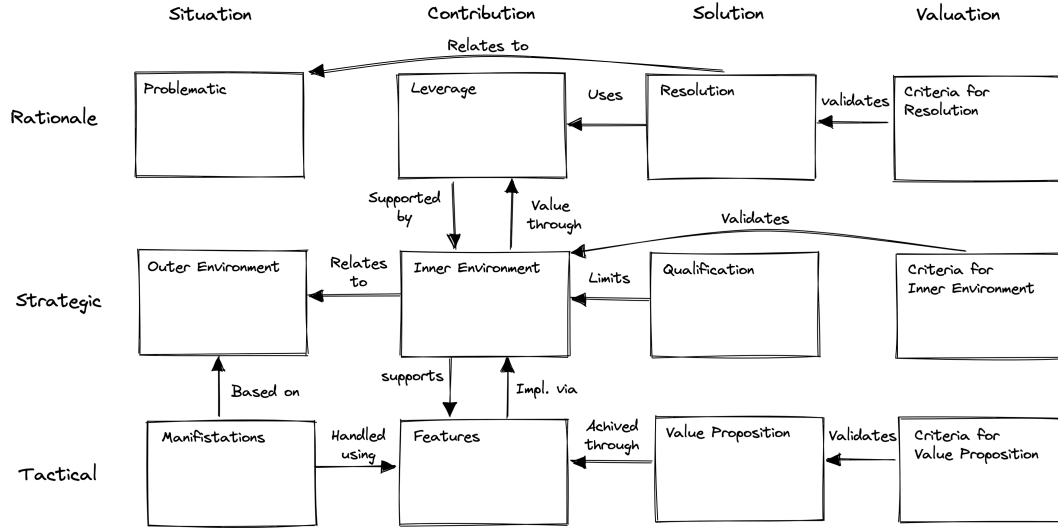


Figure 2.2: Configuration Table with cell relationships

2.6 Review in Essence

In Essence, reviews are used to verify and validate our assumptions about the problem and solution as we progress in the project. We use reviews to determine if a change is needed in the Configuration Table. Here we measure if our understanding of the problem and contribution is consistent, meaning if the contributions contribute to a reasonable solution to the problem.

In Essence, a review is done in the three focus areas: Rationale, Strategy, and Tactics. The reviews done within Essence are called *RST Reviews*.

2.6.1 RST Reviews

Essence reviews can extend any development method, whether an agile or waterfall-like development process, with additional reviews. To allow for a review model for both approaches, the model is built on a reinterpretation of the ETVX Model[1, p. 124].

The ETVX model describes software processes (agile or waterfall-like) as a set of specific activities, each consisting of entry criteria, tasks, validation, and exit criteria. The four parts of the model are described as follows:

Entry criteria (Precondition) - A criteria that should be satisfied before beginning the task.

Task - This is what we want to accomplish.

Validation (Invariant) - The procedures we use to ensure the quality of what is accomplished by the task.

eXit criteria (Postcondition) - A criteria that should be satisfied for the activity to terminate.

This model ensures that if we start in a valid state (the precondition), and if the result of the task is validated (invariant), then we can exit the model in a valid state (the postcondition).

2.6.2 ETVX Models in Essence

The precondition of the model corresponds to what we know about the problem, e.g., the situation, and where the problem manifests itself, e.g., manifestations. Knowing something about the problem and its manifestations means we are in a meaningful state. The solution we propose and the value propositions of our Prospect are mapped to the postcondition. The contribution corresponds to the tasks we carry out. If the produced outcomes of the tasks are correct per the assumptions about the problem and the solution, it will make sure we exited with a meaningful postcondition. Both the pre- and postcondition help us decide whether the task we do in the project preserves consistency between the Problem and Solution. If this is not the case, we should change the Configuration Table to restore consistency.

In Essence, the Configuration Table cells can also be interpreted as individual ETVX processes. Each cell is a structure that includes its process and internal logic, meaning that to understand, e.g., the manifestations of a problem, we start with a precondition and apply some activities to gain more knowledge and hopefully end up in a valid postcondition.

In addition, the ETVX cells can be combined in a linear process where each activity in a development process, such as agile development, where review is done after each sprint. For more waterfall-like processes, an ETVX cell can be used as a *supercell* for the whole project with individual sequential ETVX cells inside the contribution part of the supercell. The ETVX model and RST reviews can support incremental and sequential development processes.

2.6.3 Review procedure

The understanding of the problem and the solution changes over time. Changes throughout the project can be due to unforeseen circumstances affecting our digital innovation project, such as market or industry technology changes. The purpose of an RST review in Essence is to ensure the internal validity of the Configuration Table and restore potential inconsistencies. The review procedure of Essence is loosely defined as very general recommendations regarding the following steps.

Purpose. In RST Reviews, we want to identify possibilities and problems we need to consider when planning.

Prerequisites. (1) The contribution should be reviewed, e.g., when a sprint is finished, we consider what was done and when the outcome was. (2) The Configuration Table in the state at the beginning of the activity. (3) Consideration of what is essential at the end of the sprint. Moreover, an updated set of criteria based on the Valuation column in the Configuration Table.

Timing. RST Reviews should be done between every major activity in a project, such as a sprint or between the phases in more plan-driven approaches.

Participants. Anyone can be invited to participate in the RST review, but the Responders need to showcase the contributions of the sprint, and the anchor should lead the review. The Challenger and other customers and stakeholders should participate in the RST review.

Procedure. In Essence, the following steps are the procedure of RST review and are meant to be simple without too much detail[1, p. 129].

- Demonstrate the Contributions and receive feedback based on the review criteria.
- Challengers should give their feedback first.
- No blaming. Comments are noted but not up for discussion.
- Those assuming the Child role leave decisions to the team and challengers.
- At the end of a phase or sprint, the Configuration Table is updated accordingly based on feedback from the review.

2.6.4 Pivot or persevere

In Essence, we base critical decisions in the project on the RST Review. We want to determine if we continue following the current direction in the project or if we should pivot in a different direction and start working on another prospect. Essence describes several types of pivots inspired by Eric Ries' The Lean Startup[12]. The pivots, in Essence, are:

- **Zoom-in pivot** and **Zoom-out pivot**, where the contribution is narrowed or widened to include fewer or more features. The affected cells in the Configuration Table are Inner Environment, Manifestations, and Value Propositions.
- **Leverage pivot**, where we find new leverage points to use in the project. It could be new insights into the Inner Environment, Features, Value Propositions, or Qualifications.

- **Manifestation pivot**, where users are the same, the project's focus might have changed. When the Manifestations change, likely the Features, Inner Environment, and Value Propositions of the project also change.

Other pivot types exist in Essence, such as Outer Environment and Platform pivots, where new information is found in the Inner or Outer Environment.

3 | Visual Inquiry Tools

A visual inquiry tool uses generic text and images to start a conversation about and inquiry into problems. Visual inquiry tools assist in the exploration and ideation of specific problems. In [3], they propose a design theory for developing new visual inquiry tools. The authors focus on strategic management problems. They state that current tools and models need to be improved regarding how rigorously and theoretically sound the tools are designed. Visual inquiry tools are often used to enable **joint inquiry**, which is

"a process through which a group of diverse individuals who face an uncertain situation jointly define and explore a problem, and jointly generate and evaluate different hypotheses about how to solve it"[3]

They developed a design theory based on the design science research (DSR) projects that resulted in the design of Business Model Canvas[9] (BMC), Value Proposition Canvas[10] (VPC), and the Team Alignment Map[15] (TAM). Authors indicate that such tools have three main pillars: (1) a conceptual model that frames and defines the strategic management problem, (2) shared and visual design spaces used to explore and try different solutions, and (3) a set of directions for use that users can follow to solve the strategic management problem.[3]

They theorize how design knowledge can be derived from multiple projects instead of just one. Since different tools are used to solve problems, the authors investigate how to accumulate knowledge from multiple projects in design science research. For each tool, a within-project analysis was done to accumulate design knowledge. Then cross-project analyses comparing the design knowledge acquired in the three projects. The design knowledge of each project consists of the design intuitions, design decisions, principles of form and function, and descriptive knowledge used to understand the problems and design the corresponding solutions in each case.[3]

3.1 Design Principles for a Visual Inquiry Tool

The paper presents three general design principles for designing a Visual Inquiry Tool. Each of the three principles contains a series of sub-principles.

3.1.1 Design Principle 1: The conceptual model

To structure the problem which the Visual Inquiry Tool aids in solving, frame it in a conceptual model, describing the building blocks of the problem.

Frame The conceptual model should identify the components which should be inquired into to address the problem. Components should be mutually exclusive and collectively exhaustive.

Rigor & Relevance The conceptual model should be based on academic justification knowledge. Designers should validate the model with practitioners to ensure validity.

Parsimony The conceptual model should be usable for cross-boundary teams. The number of components within the conceptual model should be minimal to avoid information overload and prevent a prohibitive level of detail.

3.1.2 Design Principle 2: Shared visualization

The conceptual model should be visually represented, where the components are structured logically to facilitate communication.

Functionality The components of the conceptual model should allow for ideation, prototyping, and presentation. This is achieved by presenting the components as empty problem spaces.

Arrangement The components should be arranged according to their relationships to increase affordance. The relationships should, however, be hidden to reduce complexity.

Facilitation To further increase the affordance of the tool, visuals, such as images and metaphors, should be used.

3.1.3 Design Principle 3: Directions for use

To support a VIT, techniques allowing for joint inquiry should be specified. The criteria for the directions of use are specified as follows:

Ideation Should guide the creation and help the team exchange ideas.

Prototyping Should support the team in developing and evaluating alternative hypotheses.

Presentation Should create tangible marks, useful to present and discuss the design or solution.

3.2 Summary

In this chapter, we introduced the Design Theory of Visual Inquiry Tools. We outlined the three overarching design principles and their corresponding sub-principles, covering *the conceptual model*, *shared visualization*, and *directions for use*. These principles can be used to develop new Visual Inquiry Tools and evaluate existing ones. In the following chapter, we apply these principles to assess the Configuration Table as a Visual Inquiry Tool to gain further insight into the performance of its current format.

4 | Essence Configuration Table as a Visual Inquiry Tool

This chapter reviews the Configuration Table as a visual inquiry tool following the design principles described in section 3.1. The review aims to determine how well the Configuration Table aligns with the design principles. This review is based on our findings when working with the Configuration Table in Essence.

4.1 Review of Essence Configuration Table

To review the Configuration Table based on the design principles for visual inquiry tools, each design principle is evaluated if and how the Configuration Table fulfills it.

The Conceptual Model

Frame The Configuration Table is a conceptual model that consists of a table with columns representing the views in Essence and rows representing different levels of concern. The table's cells are defined by a title, representing a question and a view required for inquiry into a specific aspect of the problem or solution.

Rigor & Relevance The Essence methodology is based on the principles of pragmatism, and the Configuration Table is inspired by various influential models, including Toulmin's model of argumentation and elements from software development processes, such as validation criteria, value propositions, and features. The Configuration Table has been employed by students enrolled in the Software Innovation course at Aalborg University multiple times, demonstrating its rigor and practical relevance.

Parsimony The Configuration Table is a comprehensive tool that considers all aspects of a project, resulting in a large number of cells. The table can be overwhelming and potentially lead to information overload. The density of the table may obscure meaningful relationships between elements within the different cells. However, by breaking down the table and examining individual cells or patterns described in

section 2.5, it may be more manageable and increase the outcome of collaborative inquiry.

Shared Visualisation

Functionality The components (cells) of the Configuration Table is presented as empty problem spaces, defined by their position in the table. The format of the Configuration Table enables the depiction of existing knowledge about the problem and solution spaces and the discovery of new knowledge, as each cell represents a question to be answered. The user is not limited by the format when filling out the cells, but the process can be complex as the cells interact and depend on each other. Some cells, such as Problematic, Resolution, and Qualification, are further divided into "sub-cells" that address specific aspects such as Challenge and Problem, Prospect, Warrant, and Backing. The Configuration Table supports the activities of ideation, prototyping, and presentation, but opportunities for improvement exist in all of these areas.

Arrangement The components in the Configuration Table are placed in a logical order. The order is according to their relationship and placement in the problem or solution space. However, the cells have horizontal and vertical relationships, and the vertical is not intuitive unless the users follow the patterns presented in Figure 2.1. Following these patterns may require the users to consult with the theory described in Essence while filling out the Configuration Table.

Facilitation In the Configuration Table, each column is identified with a color based on the view in Essence, and the rows represent different levels of concern. These elements provide the users with a common language to use. When working with the Configuration Table, these elements aid in scoping the joint inquiry, ensuring the users are aligned on the part of the project they are discussing.

Directions for Use

Ideation There needs to be a clear direction or guidance for filling out the Configuration Table. The Configuration Table can be filled out vertically, horizontally, or unsystematically. Essence already presents a series of patterns, described in Figure 2.1, which aims to guide the team to explore different aspects, and leverage points or ideas within the problem and solution space. However, as these patterns are not explicitly illustrated on the Configuration Table itself, it can result in missing opportunities for realizing how certain elements within the problem and solution space depend on each other.

Prototyping The Configuration Table does not facilitate the comparison of alternative solutions to the problem in a structured manner. One way to address this is to create a separate Configuration Table for each alternative solution (prospect). However, this can be time-consuming due to the complexity of the process. The

users can describe alternative solutions within the same Configuration Table, but it would only increase the difficulty of distinguishing these solutions.

Presentation The Configuration Table serves as a summary of existing knowledge about the problem and solution spaces and also invites the user to explore new knowledge, all within a single table. However, the depiction of general knowledge is only sometimes effective, as the relationships between elements within the cells need to be demonstrated. For example, a Configuration Table may contain multiple value propositions. However, the relationships between these value propositions and their corresponding features may need to be clarified, making it difficult for the user to distinguish which features are associated with which value proposition. This issue may result in dangling information, where related elements are not examined when the Configuration Table is updated, leading to an inaccurate representation of the problem and solution spaces.

4.2 Summary

To some extent, the Configuration Table fulfills the Principles proposed in [3]. It enables joint inquiry into the problem and solution spaces and supports the depiction and exploration of new knowledge. However, the table's design limits its capabilities of successfully depicting the knowledge. The Configuration Table patterns allow the exploration of the interconnectedness of knowledge within the problem and solution space, but these relations are made explicitly clear to the user. This limitation may lead to reduced outcomes from joint inquiry and make it challenging to implement modifications. Additionally, evaluating different hypotheses and considering alternative solutions within a reasonable timeframe may be difficult, as this would require completing a Configuration Table for each prospect scenario.

5 | System Design

This chapter describes the system design considerations for developing Digital Essence. Firstly, we explain computer-aided software engineering (CASE) tool theory and three philosophies for creating CASE tools and how they apply to the strict modeling of systems. Secondly, we apply them in the context of Digital Essence and how they can be used in our design. Thirdly, we describe Digital Essence as a methodology companion and how methodology support is implemented for filling out the Configuration Table.

5.1 CASE Tools as Methodology Companions

A computer-aided software engineering (CASE) tool can assist in software development, management, and maintenance[7]. In [21], CASE tools are described as Methodology Companions meaning tools that assist in following a methodology (a systematic approach). A CASE tool must provide some assistance in the form of warnings or error messages if the user violates the rules or structures in the methodology. The CASE tool should either prevent the user from moving forward to the next step or provide helpful guidance to the users. In [21], this type of implementation of rules and structures in a CASE tool is called methodology support. Methodology support can be categorized into:

- (1) automated control, or
- (2) informative feedback on methodology violations

These types of methodology support are relevant for methodologies that have a way of measuring consistency in their models. In [21], the authors consider structured analysis (SA) and structured design (SD). Here it is possible to check for internal inconsistencies like all data flows have a start and an end, and all flows have names, or the users follow a top-down approach for decomposing the system. These measures can be used to verify that users are correctly using the methodology and give feedback to the users.

There are two primary aspects of software engineering where CASE tools can provide feedback and assert control:

- (1) with a general process or an approach to using a technique in systems development
- (2) and/or the output of each technique or phase in systems development.

In general [21] describes three philosophies for methodology support. Firstly, the *Restrictive* philosophy is how a system restricts the users' decision-making process to a subset of all possible processes, limiting the user's choices strictly following the methodology. An example of a CASE tool for SA/SD in software development, we can restrict the users to start modeling the whole system and decompose it in a top-down fashion. Here the users are not allowed to break the rules of model creation.

The *Guided* philosophy is how a system guides its users in constructing and executing decision-making processes by assisting them in choosing the next step. Considering the same example with SA and SD, in a guided system, a top-down approach is not strictly enforced but is suggested as a way to go about the task. No particular starting point is presented, and the CASE tool serves as a guide through model creation.

The *Flexible* philosophy gives the users the most freedom. In the same example above, the users can choose any approach, either top-down or bottom-up. In the flexible philosophy, the user can be warned that they do not adhere to the methodology, but this is never enforced. This philosophy is closely related to tools that start with a blank canvas, where it is up to the user to follow the methodology. Each philosophy differs in how they implement the methodology support and what and when they control the user's choices and serves as three general ways of implementing methodology support in CASE tools.

5.2 Essence Configuration Table as a CASE Tool

The three design philosophies for CASE tools as methodology companions work well for methodologies that use a more rigid model definition where internal consistencies can be verified. Here we can define a correct or consistent model in terms of specific requirements that need to be met. Still, for the Essence Configuration Table, no specific requirements exist for filling out the cells, and it is a far less restrictive model without set rules. Essence provides structure rather than a sequence of steps and rules.

5.2.1 Methodology Support Philosophy for Essence

The 'restrictive' philosophy is difficult to use with less strict rigid model creation methodologies since consistency check is not as straightforward without rules or sequences. The Essence Configuration Table uses structures rather than sequences to fill it out. Hence the restrictive philosophy does not work as well with the configuration table. The 'flexible' philosophy is more in line with the Essence methodology than the restrictive one, as the method provides the users with structures from which

they can build their software innovation projects. Still, in this project, the main contribution is to establish what types of help a digital version of the configuration table can provide, so following the flexible philosophy does not contribute to this purpose. Lastly, the 'guided' philosophy suggests a mix of guidance and freedom in the CASE tool. Some studies suggest that in a computer-assisted learning scenario, the 'guided' philosophy works well in two cases: (1) for tasks that involve problem-solving and decision-making and (2) for users or students with prior knowledge of the subject [21, p. 92]. These scenarios work well with this research project as the CASE tools will primarily be used by students who work with software innovation.

To summarize the choice of methodology support for Digital Essence, the 'guided' philosophy fits well with the current goals of this project, namely to introduce more guidance into the Essence configuration table by providing both a digital and a more advice-giving version.

5.3 Design of Guidance in Digital Essence

Since the Essence Configuration Table consists of structures to reason about the problem, contribution, solution, and valuation of a software innovation project, its users are highly unrestricted. The general nature of the Configuration Table, in Essence, is that it is used to gain new knowledge and facilitate conversations about the project. This section describes the guiding and advice-giving devices relevant to implementing 'guided' methodology support in the Essence Configuration Table.

5.3.1 Configuration Table Patterns

As described in section 2.5, some patterns exist in Essence, more specifically, the connections between cells. These connections are described as relationships between the different areas of the project. An example is the contribution column's relationship between the Leverage Points and the Inner Environment. Here the connections represent a relationship between the content within the cells where Leverage Points are 'supported by' the Inner Environment. These relationships are not visually depicted in the current format of Essence but can potentially help the users relate the different parts of the project with one another. The patterns depicted in Figure 2.2 of the Essence Configuration Table can serve as a guide for the users of Digital Essence. Implementing support for these connections in Digital Essence may help the users think and discuss how the project's parts tie together.

5.3.2 Restricting text in cells

As explained in section 3.1, a visual inquiry tool should support joint inquiry and the design principles of ideation, prototyping, and presentation. The ideas must be written as short and precise as possible to create a tool that supports the users in developing and evaluating alternative hypotheses, facilitating more communication between the users. To support keeping the ideas shortly, the volume of text the users

can write in the cells in Digital Essence can be limited. It can also be reduced by introducing warnings to inform the users if the text can be reduced.

5.3.3 Relationships between bullet points

As described in section 2.4, the primary purpose of the Configuration Table is to provide an overview of the project and serve as the "ground truth" for conducting RST Reviews. However, it is also used to generate new ideas and thoughts through interaction with existing knowledge about the project and conversations with relevant stakeholders. To enhance the understanding of the project, a potential improvement could be to extend the idea of showing relationships between cells, as outlined in the previous section. Furthermore, including how bullet points are related could improve the clarity of the information's interconnectedness and help reduce the risk of "dangling" information, as described in section 4.1.

5.3.4 Guiding Documentation

One limitation of the Configuration Table, as noted in section 4.1, is that users may need to refer to the underlying theory described in Essence to follow the patterns of the table and fully understand the questions posed by each cell. This context-switching may reduce the effectiveness of joint inquiry. One potential solution could be to include guiding documentation within Digital Essence that provides information about the different cells in the exact location where the Configuration Table is being completed. This would enable users to access relevant information without additional context switching.

5.4 Summary

This chapter discussed various system design considerations for creating Digital Essence. These considerations include using CASE tool theory, specifically restrictive, guided, and flexible options. After consideration, it was determined that the guided philosophy aligns most closely with the goals of Digital Essence. Three potential guidance methodology supports were proposed to implement the 'guided' philosophy. These are as follows:

- Connections between cells: These connections can guide navigating the table and connecting different project elements.
- Restricting text in cells to spark reflection and conversation: These bullet points can provide prompts for discussion and help users think more deeply about different aspects of the project.
- Connections between bullet points: Visualizing how bullet points relate to each other may improve the users' abilities to understand and comprehend the interconnectedness of the information. In addition, it helps reduce the

possibility of "dangling" information when performing changes in information within the table.

- Guiding documentation: This can assist users in fully understanding the questions posed by each cell of the Configuration Table, enabling them to focus on the joint inquiry rather than searching for necessary information within Essence. This can help to reduce the need for context switching and improve the effectiveness of the inquiry process.

6 | Theory Building

This chapter first explores our approaches to acquiring knowledge about a system's problem-solving capabilities to establish a theoretical basis for the methods applied in this project. These approaches include Naur's Theory Building View and Schon's concepts of "see-move-see" and designing as a reflective conversation. Second, it outlines the plan to employ these approaches in developing Digital Essence.

6.1 Programming as Theory Building

In [8], Naur's Theory Building View of programming suggests that the primary goal of programming is not just to create a program but to construct a theoretical understanding of how problems can be solved through program execution. This approach considers that program modifications are often necessary as the needs of a system may change. According to Naur, viewing programming as a process of constructing a theory makes it possible to make design decisions that effectively solve problems and adapt to future modifications by understanding the problem and how it is addressed through the program. In this project, we adopt the Theory Building View and consider programming as a means to develop a theoretical understanding of the Configuration Table used. To understand which features are essential for a digital version, we first need a first version of the design described in chapter 5. This design is based on our experience working with Essence and CASE tool theory and is used as initial assumptions of how Digital Essence ought to work. The first version of Digital Essence will be the basis for our theory building of how a digital version is used. Since these are for now assumptions, we also recognize that program modifications may be required in the future and strive to understand the problem and its solution sincerely to accommodate these changes effectively. These future modifications must be supported as part of the architecture and components we built in the first version.

6.2 Designing as a Reflective Conversation

Another similar approach to Naur's Programming as Theory Building is Schön's concept of designing as a reflective conversation which emphasizes the importance of ongoing reflection and dialogue in the design process [14]. This approach involves

continuously reflecting on and discussing the design process by treating the process of designing as a stepping stone for further design. Only in the act of what Schön calls a "move experiment" can we make discoveries about problems and sometimes ideas we would otherwise miss. We can only get a meaningful understanding of the situation when we "see" what we have designed. According to Schön, we often identify more in a problem or situation due to our actions rather than thinking about the problem beforehand. These ideas are formulated in Schön's "see-move-see" approach [13] consisting of observing (seeing) and gathering data about a problem or a problematic situation, taking action (moving) based on that data, reflecting on the results, and collecting new data. These approaches encourage reflection and learning to improve designs and problem-solving and can lead to more effective and adaptable designs.

Schön's concept of designing as a reflective conversation and Naur's Theory Building View of programming emphasize the importance of dialogue in the design process. These approaches recognize that the design process is not a linear or one-time event but rather an ongoing process that requires continuing reflection and engaging with the problematic situation to produce effective and adaptable solutions. In Naur's view, programming is constructing a theoretical understanding of how problems can be solved through program execution. Here we reflect on the problem-solving process, considering multiple approaches and adapting the program based on feedback and new insights, meaning a conversation between the problem and the solution. Both approaches point to actions as a fundamental step to gaining insight into a problem. Here we, as designers and programmers, have assumptions about how a problem space is situated, which we use to take action and build theories about the problem and the solution.

6.3 Theory Building for Digital Essence

The system design considerations outlined in chapter 5 are grounded in CASE tool theory and the "guided" philosophy, which emphasizes the use of methodology support as a means of guiding and advice-giving devices. For Digital Essence, this includes the implementation of the features, visible connections between cells, text restrictions in the cells, and an overview of the content of all cells. These serve as our first "move" or our first reflection *in* action as we work to determine how a digital version of the Essence Configuration Table can be effectively utilized. To accompany "see" and reflection *on* action, our development of Digital Essence follows an experimental approach. In this approach, we, as developers, seek to establish a "theory" on how Digital Essence addresses the research questions through the creation and evaluation of various features, as described by Schön's concept of designing as a reflective conversation. Creating a theory on the problem-solving capabilities of Digital Essence is necessary to effectively accommodate potential changes that may arise following the execution of the experiment described in chapter 8.

To evaluate Digital Essence throughout the project, we have used the Configu-

ration Table in Digital Essence for this project itself. Here we have used the current state of Digital Essence to gain knowledge about Digital Essence. Here we gained insight into both the usability aspects of Digital Essence and made an RST review based on the input. Here we found that for students, our current prospect is in line with the solution and criteria we set for Digital Essence, but if the tool needs to be commercialized for business, some shortcomings need to be addressed.

6.4 Summary

Our main viewpoint in this project is that the programming and design processes should be viewed as ongoing, reflective conversations that involve constructing a theoretical understanding of how problems can be solved through program execution. This approach involves continuously reflecting on and discussing the design process to adapt and improve designs. The development of the Digital Essence system follows an experimental approach in which the designers and programmers seek to establish a theory on how the system addresses research questions by creating and evaluating how Digital Essence will be used. To accommodate experiments, it is necessary also to handle potential changes that may arise following the execution of the experiment, and this should be considered as a part of Digital Essence's architecture.

7 | First Version

This chapter introduces the first version of Digital Essence, including the user interface and the technologies used to develop it. The implemented features presented in this chapter are based on the four methodology companions described in section 5.3.

7.1 User Interface

The user interface for Digital Essence comprises two pages: a home page, which allows the user to manage existing workspaces and create new ones, and a workspace page which includes the Configuration Table. A workspace symbolizes a prospect, whereas the Configuration Table is used to explore and depict knowledge about the problem and solution space for the given prospect.

7.1.1 Home Page

Digital Essence enables users to create and manage multiple workspaces accessible from the home page. As shown in Figure 7.1, the home page allows users to navigate between existing workspaces and create new ones.

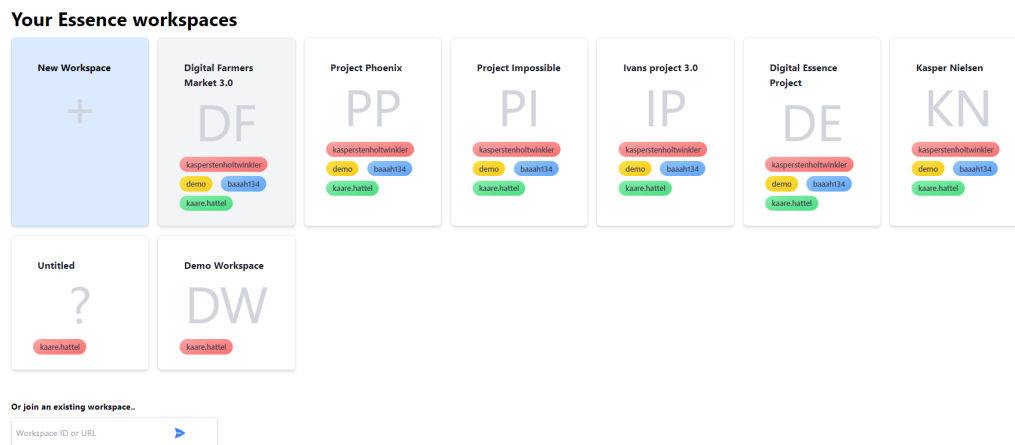


Figure 7.1: Home page

7.1.2 Workspace Page

The workspace page, illustrated in Figure 7.2, consists of a Configuration Table and a wiki detailing information regarding the different cells. Furthermore, users can be invited to a workspace for collaboration.

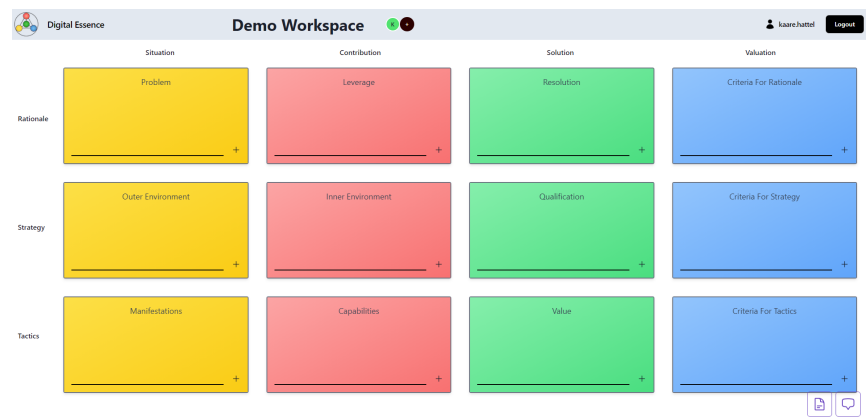


Figure 7.2: Workspace page

Configuration Table

When a user interacts with the Configuration Table by clicking on a cell, the table displays the relation between the selected cell and related cells. In addition, cells that are not to be evaluated are grayed out. These changes can be observed in Figure 7.3, where the Resolution cell is selected. The relationships are depicted by arrows and correspond to the patterns outlined in Section 2.5. The graying out of cells serves to aid the user in focusing on the selected cell and its relationships, as discussed in Section 4.1 (Parsimony).

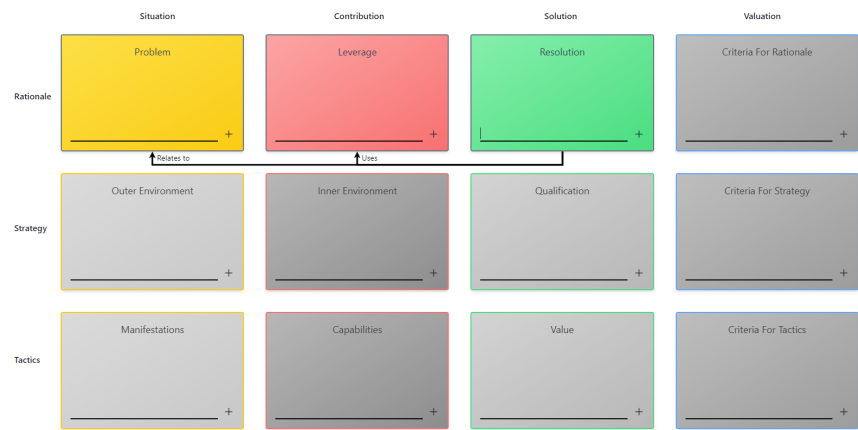


Figure 7.3: Interaction with the Configuration Table

Relations between Bullet Points

As mentioned in Section 4.1 (Presentation), one criticism of the original Configuration Table, in Essence, is the lack of ability to relate bullet points to one another. Digital Essence addresses this issue by enabling users to establish relationships between bullet points. An example of this can be seen in Figure 7.4, where the value proposition "Anticipation and detection of acute depression" is related to the two capabilities "Digital diary" and "Trend analysis," denoted by the link icon. The example is based on the Psyche case provided in the Essence textbook [1].

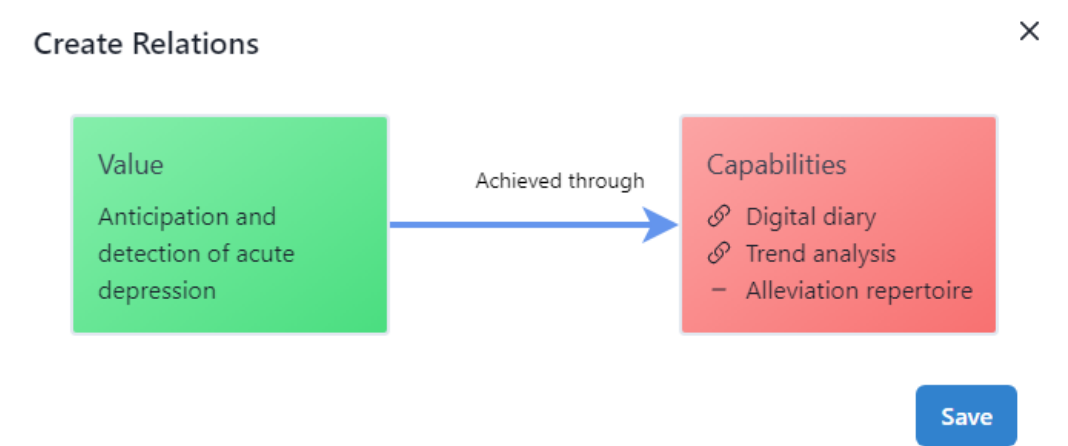


Figure 7.4: Modal to build relations between bullet points

Once relationships between bullet points have been established, they can be highlighted in the Configuration Table, as shown in Figure 7.5. In this example, the relationship created in the previous example is highlighted.

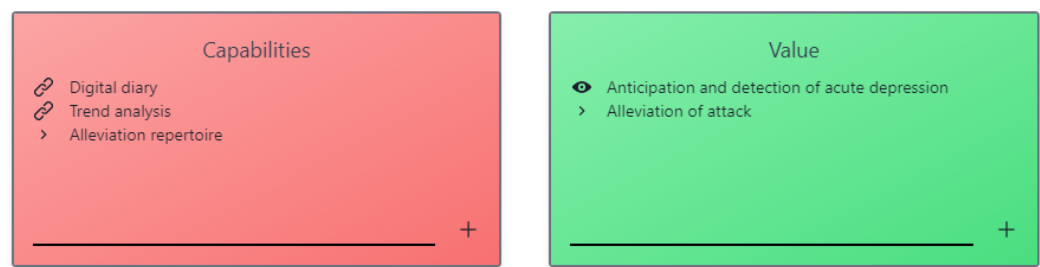


Figure 7.5: Modal to build relations between bullet points

Wiki

The wiki feature in Digital Essence is accessible as an overlay to the workspace page through the Wiki button in the bottom right corner, as shown in Figure 7.6. A user

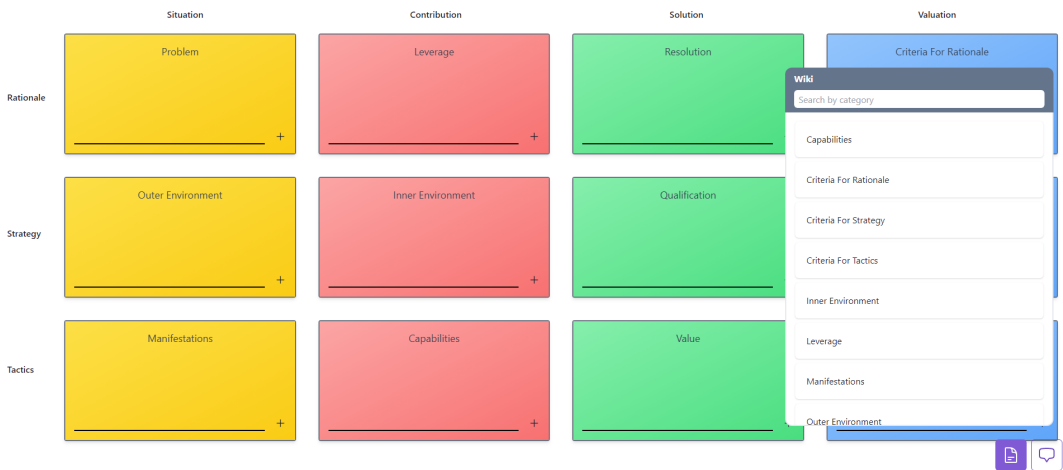


Figure 7.6: Wiki overlay

can navigate to the documentation of a specific cell through the search functionality or by selecting a cell title from the wiki. Upon selection, the user is presented with a detailed description of the chosen cell, as illustrated in Figure 7.7.

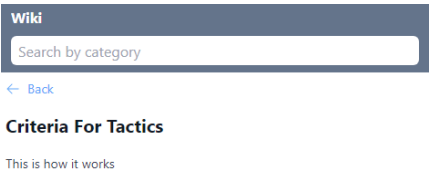


Figure 7.7: Wiki description example

7.2 Implementation

This section outlines the software architectural decisions and the technologies employed in developing the first version of Digital Essence. Here we explain the different parts of the system and how they are connected.

The first version of Digital Essence is built based on our initial assumptions about how a digital version should work. chapter 5 and chapter 6 outline this project’s approach to developing a theory about using Digital Essence. Here we established that we needed to create a first version to gain insight into how the system will be used. We need to act (move) to learn (see) what effect it has. With this approach, we will learn more about the problem and adjust the system accordingly. To work effectively with programming as theory building and to accommodate future changes to the system, we need an architecture and components that can handle change. In this project, we choose to work with the "T3-stack,"[16] which comprises a frontend,

a backend, and a database. Both the front and backend are implemented in the Node.js ecosystem. This selection of software development technologies (the stack) focuses on simplicity and modularity to adapt to changes.

The technologies used in Digital Essence are as follows:

React [4] is an open-source JavaScript library for building user interfaces. It allows developers to create reusable UI components and efficiently update the UI in response to user interactions.

Next.js [20] is a framework built on React that enables developers to build server-rendered React applications. It handles server-side rendering, code splitting, and optimized performance out of the box.

tRPC [19] is a framework for building end-to-end, typesafe API's. It allows server-side changes to be present in the frontend, enabling real-time updates to the user interface.

Vercel [18] is a cloud platform for hosting and deploying web applications. It provides a way to deploy and manage applications, with built-in support for popular frameworks such as Next.js.

Prisma [11] is an open-source database toolkit that provides a way to access and manipulate data stored in a database. The database queries are written in GraphQL, an open-source query language.

Auth0 [2] is a platform offering authentication and authorization services to manage access to applications.

Digital Essence's architecture is closely related to the Model-View-Controller (MVC) architectural pattern, where the model represents the data and logic of the application, the view is the user interface (UI), and the controller (API) handles the communication between the model and the view. Figure 7.8 is a model of where each of the technologies is used with each other in Digital Essence. Vercel hosts our Next.js application using a serverless architecture, where incoming requests are handled by serverless functions, which communicate with the MSSQL database hosted on our server and our Auth0 instance.

The database is structured according to the entity relations diagram in Figure 7.9¹. The entities are structured to allow us to make changes without invalidating existing data, such as changing the titles and relationships between cells. When a user creates a new workspace, this data is stored in the **Workspace** entity. It includes a title and a unique Id and is represented as the **Workspace** entity. Each workspace can have Cells related to them. The workspaces have twelve cells assigned to them when created, constituting a Configuration Table. The data for each cell is kept in

¹Greyed-out entities are used for user management via Auth0 and will not be considered in this section.

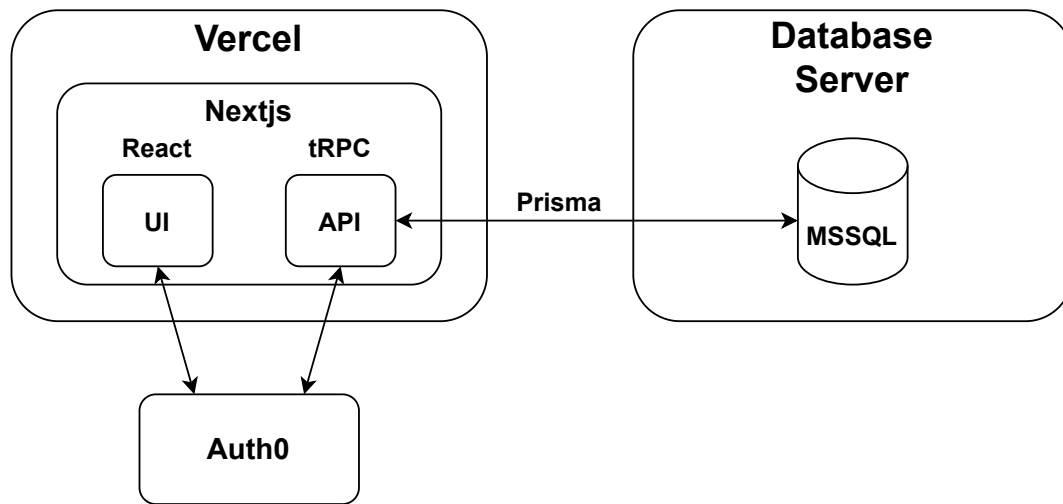


Figure 7.8: Model of where the technologies are used in Digital Essence

the **Cell** entity, which refers to a **Category** entity with the cell's position and title. The categories have relations to one another through the entity **CategoryRelations** that keep track of the direction of the relationship and the description. Similarly, the **Cell** entity can have multiple **CellBullet** entities referencing it. *Relationships* between CellBullets is defined as a **BulletRelation** entity, which also contains a direction (from and to).

The **Audit** entity can capture changes in all of the abovementioned entities. It can capture changes for each user and its workspaces. With the **Audit** entity, we can capture what entity changes and when and what the old and new values are. The Audits will be used to facilitate the experiments of Digital Essence.



Figure 7.9: Entity Relation Diagram of Digital Essence

8 | Experiment

This chapter presents the experimental design for the studies conducted during the 10th semester. Further, primary objectives and intended usage of the results will be outlined, as well as the target groups on which the experiments will be conducted. Additionally, we will describe the data-gathering mechanism implemented in the Digital Essence system to ensure that data collected during the experiments is accurately recorded and stored.

8.1 Experiment

This study aims to investigate the potential benefits of digitizing the Configuration Table and understand how it is being used by practitioners, through an experiment involving 8th-semester students in the spring semester of 2023, following the Software Innovation course by Ivan Aaen.

The experiment consists of two parts: asking students to use Digital Essence when doing their mini-project assignment in the Software Innovation course. The other includes interviews to gain subjective feedback and further support the data collected from the first part of the experiment.

The experiment aims to validate the selected features for a digital Configuration Table in response to RQ2 and answer RQ3. We would like, in particular, to investigate the following:

In previous chapters, we have established the essential features for the digital version of the Configuration Table as four methodology companions. These features are selected based on the analysis of the Configuration Table as a VIT, CASE tool theory, and the adopted theory-building methods. We set out to confirm those features' usefulness by answering RQ3. We can in details investigate the following:

- Does the configuration table patterns impact the order in which the table is being filled out or modified?
- Do practitioners establish and maintain relationships between bullet points?
- Makes the bullet point relationships impact how users create and modify the bullet points?

- Do practitioners use the documentation while filling out the Configuration Table?
- If so, how long do they spend on each documentation page?
- Are some documentation pages more important than others, based on the frequency they are being used?
- Can we predict a suitable role for each group member based on their interactions with Digital Essence?

We investigate these questions to determine if the implemented features effectively support practitioners working with the Configuration Table. We also hope to use the interviews to explore any relevant unevaluated features identified by the practitioners. The findings of this study also inform future developments in the Essence methodology and the Digital Essence platform.

8.2 System auditing

Performing the experiments should result in us being able to answer RQ2 and RQ3 regarding essential features of Digital Essence and how practitioners are using the digital configuration table. We have implemented an auditing system that captures user interactions such as:

- Creation and modification of bullet points.
- Creation and modification of bullet point relationships.
- Using the wiki documentation.
- How much time a user spends on a given page of documentation.

The information we capture from these user interactions allows us to investigate the order in which the configuration table is completed and modified and if the essential features identified in response to RQ2 are used as anticipated.

The auditing system, implemented as middleware, intercepts requests to the database and stores metadata, including user IDs, timestamps, and time duration, allowing each change to be related to a specific time, user, and workspace. The auditing system also enables us to create a historical timeline for each workspace's changes. The table storing the experiment-related data can be seen in Figure 7.9 as the "Audit" entity.

In addition, the potential benefits of implementing an analytics platform tool, such as Google Analytics or PostHog, could be explored. These tools offer insightful features such as:

- Split Testing (A/B testing): A way to randomize multiple features and their performance.

- Heat Maps: Visualize user activity identifying hot and cold areas in the Configuration Table.
- Event Tracking: General events such as mouse click, mouse over, form interactions, and image interactions.

While also providing a user-friendly interface for analyzing and visualizing the collected data. Implementing such a tool further strengthens the data used to validate the experiment by tracking additional user events and behaviors that are not necessarily predicted prior to the experiment.

9 | Discussion

In this chapter, we will discuss whether the research questions have been answered, our implementation choices, and how following Naur's Theory Building View and Schon's reflective conversation have impacted the project. Lastly, we outline future work.

9.1 Discussion of Research Questions

The research questions are as follows:

- RQ1: Is the Configuration Table in alignment with the Visual Inquiry Tool Design Theory Principles?
- RQ2: Which features are essential for a digital version of the Configuration Table?
- RQ3: How is the digital version of the Configuration Table used?

RQ1: Is the Configuration Table in alignment with the Visual Inquiry Tool Design Theory Principles?

As described in section 4.1, the Configuration Table fulfills the visual inquiry tool principles to some extent. It allows for a joint inquiry into the problem and solution and supports the exploration of new knowledge. However, its current format does not allow full utilization of the Configuration Table patterns, and the relationships between bullet points are not presented to the user. These limitations may lead to reduced outcomes of the joint inquiry and increase the difficulty of performing changes to the knowledge depicted. If changes are not done correctly, it may lead to "dangling" information, resulting in an inaccurate representation of the problem and solution spaces. In section 5.3, we proposed different solutions to accommodate the shortcomings of the Configuration Table. These include highlighting Configuration Table patterns and bullet point relationships, restricting text in cells, and providing the users with helpful information on the exact location of the Configuration Table.

RQ2: Which features are essential for a digital version of the Configuration Table?

As we adopt Naur’s perspective of Programming as Theory Building and Schön’s concept of Designing as a Reflective Conversation, the first version of Digital Essence described in chapter 7 is mainly based on our experience working with the Essence methodology. In addition, we have applied VIT design principles to evaluate the current Configuration Table in Essence. In chapter 4, we identified certain limitations in the current Configuration Table, including complexity in filling out the table and hidden elements that can inhibit ideation. We also found that the Configuration Table lacks guidance on how users interact with it. In response, we have used CASE tool theory to implement a guided philosophy in Digital Essence, introducing the methodology supports (1) connections between cells, (2) restricting text in cells to spark reflection and conversation, (3) connections between bullet points and, (4) guiding documentation. These methodology companions relate to the identified issues highlighted in the analysis of the Configuration Table as a VIT and our own experiences. These are implemented as features in Digital Essence. At the time of writing, we have yet to implement the restricted text in cells. The restriction of text should be accommodated before the experiments are to begin next semester. To fully address RQ1, it is necessary also to evaluate Digital Essence against the VIT design principles. While this has yet to be done, we plan to conduct such an evaluation to benchmark Digital Essence against these principles in future work.

RQ3: How is the digital version of the Configuration Table used?

In response to RQ2, we have implemented the methodology companions in Digital Essence described above. We anticipate these methodology companions impact how practitioners use Digital Essence. We hypothesize that by providing visual representations of the relationships between cells, users will be more likely to follow them as they complete the table, resulting in a more interconnected and accurate depiction of the problem being addressed.

Additionally, the bullet point relationships have been included to improve the clarity of how different pieces of information within the table are connected. Changes to one bullet point may require corresponding changes to other bullet points in different cells. By improving the clarity of these relationships, we aim to ensure the overall validity of the Configuration Table. To further support the validity of the information within the table, a further improvement would be to implement a guided methodology companion that alerts users to the potential need for modifying related bullet points when making changes to a particular bullet point. However, this addition still needs to be implemented, as the main focus has been on finishing how cell and bullet point relations are established in the system.

To address RQ3 more fully, we have implemented mechanisms for storing data on the completion and modification of the Configuration Table. These mechanisms, described in more detail in section 8.2, allow us to analyze the order in which the

table is being filled out and modified, as well as track how bullet points are related and how changes to one bullet point may impact others. This data will be valuable in understanding the effects of explicitly displaying cell and bullet point relationships on the completion and modification of the table.

9.2 Discussion of Implementation

Choice of Technology

The current technology stack described in section 7.2 allowed for a quick start to the project, enabling us to produce the prototype quickly. However, we encountered limitations during development, such as lacking temporal tables for data logging. One potential solution to these limitations could be switching to a backend technology such as ASP.NET (C#), which offers more robust data logging and management options.

One of the factors influencing our technology selection was the decision to utilize React and Next.js as the foundation of our technology stack. As previously mentioned in section 7.2, Next.js employs both client-side and server-side rendering, effectively merging the client-side and server-side. To ensure type safety and coherence between our database models and UI in React, we implemented tRPC, ensuring that changes in a backend model are immediately reflected in the frontend.

Auditing

It is vital to implement effective auditing measures to ensure the traceability of user actions we obtain during experiments. These are the actual results to answer our research questions. Our solution is to manage the data in a single table; this requires us to ensure that all changes are correctly logged and tracked. There are more optimal approaches than the approach described chapter 8, but it works as expected. However, changing the current data auditing implementation and extending it is time-consuming. The auditing is implemented by intercepting all requests and saving relevant data. Another potential solution would be using temporal tables, which allow for the historical tracking of data changes in the database system. Temporal tables are supported by entity framework in ASP.NET, where auditing for each database table within the system requires little to no setup.

9.3 Discussion of Theory Building

The choice to utilize the Naur and Schön frameworks for theory building in this project was motivated by their focus on the iterative, reflective process of knowledge construction. Naur's concept of "programming as theory building" and Schön's notion of "see-move-see" emphasizes the importance of ongoing reflection and learning in developing new knowledge and understanding. These approaches are particularly

well-suited to software development's dynamic, iterative nature, where knowledge is constantly evolving and refined. In this project, these frameworks provided value by allowing us to focus on developing the Digital Essence while learning about its nature and improving it. While developing Digital Essence, we also used this project within Digital Essence to evaluate and test features. This approach yielded valuable results, such as how well the structure of the elements in the user interface allows for joint inquiry and maintaining an overview.

9.4 Future Work

In this section, we outline future work which will improve the overall experience and performance of Digital Essence. These work items consist of additional metrics for data auditing, user interface improvements in terms of styling and functionality, and changes in terminology.

9.4.1 Additional Metrics

If we introduce additional metrics, we could increase the precision of our evaluation of Digital Essence performance in the experiments. In addition to gaining new insights into measure points, we otherwise would not have access to them. One way to introduce additional metrics is implementing temporal tables, as they would provide a change history of all entities. Our change history is manually configured by selecting and implementing data logging on relevant entities. In addition, temporal tables are more robust than our current solution and easier to maintain. As a framework to manage temporal tables is not supported in our current setup, we would have to implement another backend, such as one using ASP.NET (C#).

Another way of enhancing the experiment results is by implementing a framework that generates heatmaps of the various Configuration Tables. These heatmaps could provide valuable insights into user behavior and may not be easily captured through other means.

9.4.2 User Interface Improvements

In chapter 7, we introduced the current version of Digital Essence, including its user interface. Currently, the Wiki containing documentation on the cells is located in the bottom right corner of the screen and functions as an overlay. As shown in Figure 7.6, the Wiki covers the entire Validation column, which can be inconvenient for users who wish to consult the documentation while filling out cells in this column. One potential solution to this issue is to implement the Wiki as a toggleable sidebar, which shrinks the Configuration Table when expanded, ensuring that all necessary information is accessible at all times.

Additionally, based on the feedback received and the progression of the Experiment, there may be a need to reevaluate how bullet point relationships are presented.

The current version, depicted in Figure 7.5, may not be sufficiently clear or intuitive to support users effectively.

To improve joint inquiry and transparency, we want future work to improve online collaboration by showing the location of each user in a workspace. This improvement is inspired by other collaboration tools, such as Google Docs or Overleaf, where the cursor location of each user is presented.

9.4.3 Terminology and Essence related changes

Digital Essence is a tool for a joint inquiry into software innovation projects that utilizes intuitive and precise terminology to facilitate optimal outcomes. As Digital Essence is based on the Essence Configuration Table, it is appropriate to adopt the Essence terminology. Specifically, the term "Workspace" should be replaced with "Prospect" to align with the Configuration Table's depiction of a given prospect's problem and solution space. In addition, the Configuration Table employs Toulmin's model of argumentation in the Problematic, Resolution, and Qualification cells, which contain "sub-cells" as described in section 2.4. These sub-cells have yet to be implemented in Digital Essence.

10 | Conclusion

In this project, we explore the concept of a digital version of the Configuration Table, one of the Essence methodology's critical structures, to evaluate its suitability as a visual inquiry tool.

We established the following research questions:

- RQ1: Is the Configuration Table in alignment with the Visual Inquiry Tool Design Theory Principles?
- RQ2: Which features are essential for a digital version of the Configuration Table?
- RQ3: How is the digital version of the Configuration Table used?

Concerning RQ1, we analyzed the Configuration Table as a VIT and concluded that it fulfills the Principles proposed in [3] to some extent. It enables joint inquiry into the problem and solution spaces and supports the depiction and exploration of knowledge. However, the table is limited by its current format. The configuration table patterns described in section 2.5 are not depicted, nor are the relationships between bullet points. This results in the Configuration Table failing to display the interconnectedness of the knowledge in the problem and solution space and may lead to "dangling" information when the table is being modified.

We used the conclusion points from the analysis, in combination with Naur's definition of Programming as Theory Building [8] and Schön's concept of Designing as a Reflective Conversation [13] in addition to CASE Tool theory [7] to establish essential features in the digital Configuration Table and thus contribute to RQ2. These features were proposed as the following guided methodology supports (1) connections between cells, (2) restricting text in cells to spark reflection and conversation, (3) connections between bullet points, and (4) guiding documentation.

We outlined an experiment consisting of two parts to validate the essential features and determine how practitioners use the digital Configuration Table. With the experiment, we created an auditing system that stores the necessary information when users interact with Digital Essence to validate RQ2 and answer RQ3.

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